Project: Worked examples
Title: Duct sizing
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

Sizes each duct section by the equal-friction method (a target friction rate) or the velocity method (a target velocity), rounds to standard sizes, and reports velocity, friction rate, friction and fitting losses per section and the total pressure drop along the index run, with warnings against user velocity and pressure-drop limits.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Round duct diameter from the design target | D by Colebrook with the air properties at T (equal friction), or $D = \sqrt{(4Q/\pi v)}$ (velocity), rounded up to the next standard size |
| Rectangular duct of equal friction and capacity | $D_e$ = 1.30 (ab)^0.625/(a + b)^0.25 |
| Flat-oval duct of equal friction and capacity | $D_e$ = 1.55 A^0.625/P^0.25, with $A = \pi b^2/4 + b(a - b)$ and $P = \pi b + 2(a - b)$ |
| Friction loss per metre, at the velocity $v_e$ of the equivalent round duct | $\Delta p/L = f \rho v_e^2/(2D_e)$ |
| Fitting losses, at the actual velocity | $\Delta p = \Sigma C \rho v^2/2$ |
| Index-run totals, summed over every section of the run | $\Delta p_{run} = \Sigma(\Delta p/L \cdot L + \Sigma C \rho v^2/2)$ |

Basis: Duct design (friction chart basis, equivalent diameters, loss coefficients)[1](Ch. 21); Flow of fluids in pipes and ducts[2](Section 4); Ventilation for buildings: sheet metal air ducts and fittings with rectangular cross-section, dimensions[3]; Ventilation for buildings: sheet metal air ducts and fittings with circular cross-section, dimensions[4].

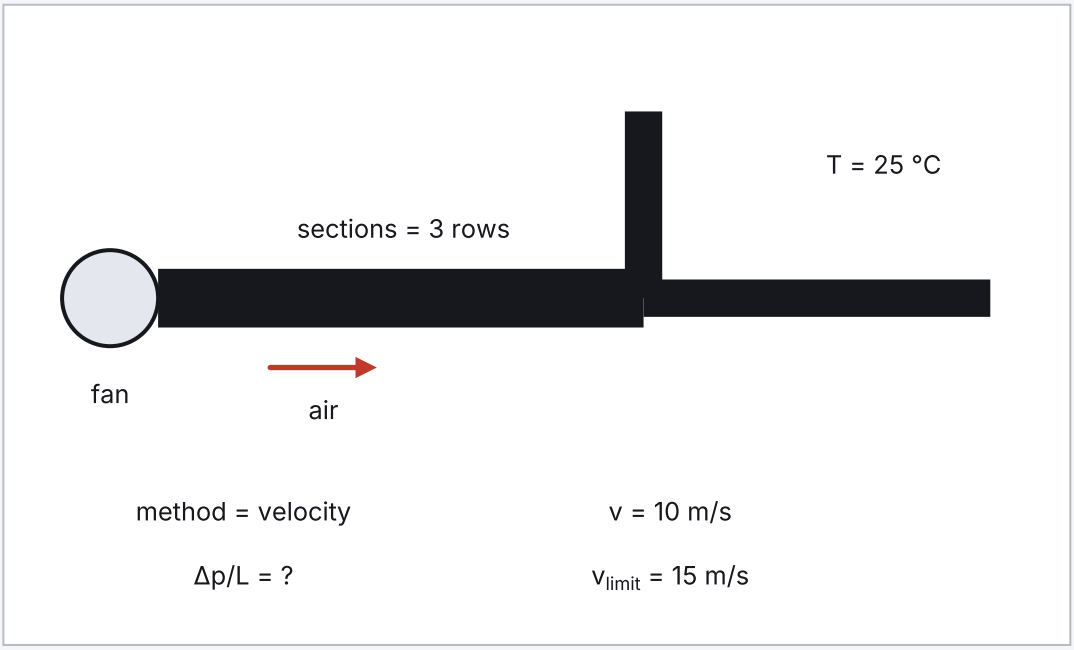



Figure 1 — A duct run from the fan through its sections as listed (a main and two branches drawn), each sized by the chosen method. Not to scale; the labels are the inputs as entered.

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| method | Sizing method | velocity | | assumption: Velocity method for an extract system. | any |
| $dp_{target}$ | Target friction rate (equal friction) | 1 | Pa/m | default | 0.3 … 5 Pa/m |
| $v_{target}$ | Target velocity (velocity method) | 10 | m/s | spec: Dust-carrying extract: 10 m/s minimum transport velocity, SPEC-M-09 | 2 … 20 m/s |
| T | Air temperature | 25 | °C | assumption: Workshop air 25 °C. | -20 … 60 °C |
| $p_{atm}$ | Barometric pressure | 100.5 | kPa | assumption: Site 70 m AOD, ≈ 100.5 kPa. | 80 … 105 kPa |
| eps | Duct roughness | 0.15 | mm | assumption: Spiral-wound galvanised duct with joints. | 0.03 … 3 mm |
| sections | Duct sections | 3 row(s) | | drawing: DRG-M-310 rev A | any |
| $v_{limit}$ | Velocity limit (warning and check) | 15 | m/s | spec: SPEC-M-09 noise limit | 2 … 30 m/s |
| $dp_{limit}$ | Total pressure-drop limit (check) | 600 | Pa | spec: Extract fan external static 600 Pa | 50 … 2000 Pa |
| $round_{to}$ | Round sizes up to a multiple of | 50 | mm | spec: BS EN 1506 preferred sizes | 1 … 100 mm |

| TABLE ROW | VALUES |
|---|---|
| sections[1] | name = hood branch; Q = 0.35 m^3/s; L = 6 m; shape = round; a = —; D = —; C = 1.5; dp_eq = 120 Pa; index = yes |
| sections[2] | name = main; Q = 1.05 m^3/s; L = 20 m; shape = round; a = —; D = —; C = 0.6; dp_eq = 0 Pa; index = yes |
| sections[3] | name = stack; Q = 1.05 m^3/s; L = 8 m; shape = round; a = —; D = —; C = 1; dp_eq = 30 Pa; index = yes |

## 3. Assumptions

A1. Air is dry at the stated temperature and pressure (ρ = p/(287.055 T)); viscosity by Sutherland's law. (Standard air properties.)
A2. Rectangular and flat-oval ducts are sized through the equivalent diameter for equal friction and capacity. (Huebscher / Heyt–Diaz relations as given in ASHRAE Fundamentals Ch. 21.)
A3. Fitting loss coefficients C are user-supplied (manufacturer or ASHRAE/CIBSE data, not embedded) and refer to the section's velocity pressure. (Non-negotiable: no standard tables embedded.)
A4. method (Sizing method) is taken as velocity. (Velocity method for an extract system.)
A5. T (Air temperature) is taken as 25 degC. (Workshop air 25 °C.)
A6. $p_{atm}$ (Barometric pressure) is taken as 100.5 kPa. (Site 70 m AOD, ≈ 100.5 kPa.)

A7. eps (Duct roughness) is taken as 0.15 mm. (Spiral-wound galvanised duct with joints.)

### 4. Working

[10] $\rho = p / (287.055\ T)$
$= 100500\ Pa / (287.055 \times 298.15\ K)$
$= 1.1743\ kg/m^3$
Air density

[11] $\mu$ = Sutherland's law
$= T = 298.15\ K$
$= 18.371\ uPa{\cdot}s$
Air dynamic viscosity

[12] $D_{s1}$ = hood branch: diameter
= target 211.1 mm rounded up to 50 mm
= 250 mm
Diameter, hood branch

[13] $v_{s1}$ = hood branch: Q / A
$= 0.35 / 0.049087$
$= 7.1301\ m/s$
Velocity, hood branch

[14] $Re_{s1}$ = hood branch: $\rho\ v_e\ D_e / \mu$ ($v_e = 4Q/\pi D_e^2$)
$= 1.1743 \times 7.1301 \times 0.25 / 0.000018371$
$= 113940$
Reynolds number, hood branch

[15] $f_{s1}$ = hood branch: friction factor
= Colebrook–White
= 0.02041
Friction factor, hood branch

[16] $dpL_{s1}$ = hood branch: $f\ \rho\ v_e^2 / (2\ D_e)$
$= 0.02041 \times 1.1743 \times 7.1301^2 / (2 \times 0.25)$
$= 2.4369\ Pa/m$
Friction rate, hood branch

[17] $dp_{f,s1}$ = hood branch: $(\Delta p/L)\ L$
$= 2.4369 \times 6$
$= 14.622\ Pa$
Friction loss, hood branch

[18] $dp_{fit,s1}$ = hood branch: $\Sigma C\ \rho\ v^2/2$
$= 1.5 \times 29.849$
$= 44.774\ Pa$
Fitting loss, hood branch

[19] $dp_{s1}$ = hood branch: friction + fittings + equipment
$= 14.622 + 44.774 + 120$
$= 179.4\ Pa$
Section loss, hood branch (index run)

[20] $D_{s2}$ = main: diameter
= target 365.64 mm rounded up to 50 mm
= 400 mm
Diameter, main

[21] $v_{s2}$ = main: Q / A
$= 1.05 / 0.12566$
$= 8.3556\ m/s$
Velocity, main

[22] $Re_{s2}$ = main: $\rho\ v_e\ D_e / \mu$ ($v_e = 4Q/\pi D_e^2$)
$= 1.1743 \times 8.3556 \times 0.4 / 0.000018371$
$= 213630$
Reynolds number, main

[23] $f_{s2}$ = main: friction factor
= Colebrook–White
= 0.018028
Friction factor, main
[24] $dpL_{s2}$ = main: f ρ $v_e^2$ / (2 $D_e$)
= 0.018028 × 1.1743 × 8.3556² / (2 × 0.4)
= 1.8475 Pa/m
Friction rate, main
[25] $dp_{f,s2}$ = main: (Δp/L) L
= 1.8475 × 20
= 36.95 Pa
Friction loss, main
[26] $dp_{fit,s2}$ = main: ΣC ρ v²/2
= 0.6 × 40.992
= 24.595 Pa
Fitting loss, main
[27] $dp_{s2}$ = main: friction + fittings + equipment
= 36.95 + 24.595 + 0
= 61.545 Pa
Section loss, main (index run)
[28] $D_{s3}$ = stack: diameter
= target 365.64 mm rounded up to 50 mm
= 400 mm
Diameter, stack
[29] $v_{s3}$ = stack: Q / A
= 1.05 / 0.12566
= 8.3556 m/s
Velocity, stack
[30] $Re_{s3}$ = stack: ρ $v_e$ $D_e$ / μ ($v_e$ = 4Q/π$D_e^2$)
= 1.1743 × 8.3556 × 0.4 / 0.000018371
= 213630
Reynolds number, stack
[31] $f_{s3}$ = stack: friction factor
= Colebrook–White
= 0.018028
Friction factor, stack
[32] $dpL_{s3}$ = stack: f ρ $v_e^2$ / (2 $D_e$)
= 0.018028 × 1.1743 × 8.3556² / (2 × 0.4)
= 1.8475 Pa/m
Friction rate, stack
[33] $dp_{f,s3}$ = stack: (Δp/L) L
= 1.8475 × 8
= 14.78 Pa
Friction loss, stack
[34] $dp_{fit,s3}$ = stack: ΣC ρ v²/2
= 1 × 40.992
= 40.992 Pa
Fitting loss, stack
[35] $dp_{s3}$ = stack: friction + fittings + equipment
= 14.78 + 40.992 + 30
= 85.772 Pa
Section loss, stack (index run)

[36] $dp_{index}$ = Σ section losses on the index run
= 3 section(s)
= 326.71 Pa
Index-run pressure drop
[37] $dp_{all}$ = Σ section losses, all sections
= 3 section(s)
= 326.71 Pa
Sum over every section (not a path)
[38] $v_{peak}$ = highest section velocity
=
= 8.3556 m/s
Highest velocity
[39] $u_{dp}$ = $dp_{index}$ / $dp_{limit}$
= 326.71 / 600
= 0.54452
Pressure-drop utilisation
[40] $u_v$ = $v_{peak}$ / $v_{limit}$
= 8.3556 / 15
= 0.55704
Velocity utilisation

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $D_{s1}$ | 250 mm | D, hood branch |
| $v_{s1}$ | 7.1301 m/s | v, hood branch |
| $Re_{s1}$ | 113940 | Re, hood branch |
| $f_{s1}$ | 0.02041 | f, hood branch |
| $dpL_{s1}$ | 2.4369 Pa/m | Δp/L, hood branch |
| $dp_{f,s1}$ | 14.622 Pa | Friction loss, hood branch |
| $dp_{fit,s1}$ | 44.774 Pa | Fitting loss, hood branch |
| $dp_{s1}$ | 179.4 Pa | Section loss, hood branch |
| $D_{s2}$ | 400 mm | D, main |
| $v_{s2}$ | 8.3556 m/s | v, main |
| $Re_{s2}$ | 213630 | Re, main |
| $f_{s2}$ | 0.018028 | f, main |
| $dpL_{s2}$ | 1.8475 Pa/m | Δp/L, main |
| $dp_{f,s2}$ | 36.95 Pa | Friction loss, main |
| $dp_{fit,s2}$ | 24.595 Pa | Fitting loss, main |
| $dp_{s2}$ | 61.545 Pa | Section loss, main |
| $D_{s3}$ | 400 mm | D, stack |
| $v_{s3}$ | 8.3556 m/s | v, stack |
| $Re_{s3}$ | 213630 | Re, stack |
| $f_{s3}$ | 0.018028 | f, stack |

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $dpL_{s3}$ | 1.8475 Pa/m | Δp/L, stack |
| $dp_{f,s3}$ | 14.78 Pa | Friction loss, stack |
| $dp_{fit,s3}$ | 40.992 Pa | Fitting loss, stack |
| $dp_{s3}$ | 85.772 Pa | Section loss, stack |
| $dp_{index}$ | 326.71 Pa | Index-run Δp |
| $dp_{all}$ | 326.71 Pa | Σ all sections |
| $v_{peak}$ | 8.3556 m/s | Highest velocity |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Velocity ≤ limit | $u_v$ = 0.55704 | $u_v$ ≤ 1 | 0.557 | |
| PASS | Index-run Δp ≤ limit | $u_{dp}$ = 0.54452 | $u_{dp}$ ≤ 1 | 0.545 | |

## 7. Applicability

WARNING — input.dp_target: $dp_{target}$ (Target friction rate (equal friction)) was not entered; the template default 1 Pa/m is used.

## 8. Verification

```
Template bs.duct-sizing@1.0.1 Content hash 25b304ddb2f458ea Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
```

## 9. References

1. American Society of Heating, Refrigerating and Air-Conditioning Engineers. ASHRAE Handbook—Fundamentals. ASHRAE.
2. Chartered Institution of Building Services Engineers. CIBSE Guide C: Reference Data. CIBSE; 2007.
3. British Standards Institution. Ventilation for Buildings—Sheet Metal Air Ducts and Fittings With Rectangular Cross-Section—Dimensions. BS EN 1505. BSI.
4. British Standards Institution. Ventilation for Buildings—Sheet Metal Air Ducts and Fittings With Circular Cross-Section—Dimensions. BS EN 1506. BSI.